RETAIL BUSINESS PERFORMANCE & PROFITABILITY ANALYSIS

Internship Project Report | Elevate Labs | Data Analyst Domain
Dataset: Synthetic Retail Transactions 2022–2023 | 2,000 Records

Total Revenue	Total Profit	Avg Margin	Total Orders	Loss Orders
■ 49.8 Lakhs	■ 17.18 Lakhs	32.21 %	2,000	34 (1.7%)

1. Introduction

Retail businesses generate massive transactional data daily. Without systematic analysis, profit-draining categories, overstocked inventory, and poor discount strategies remain invisible. This project applies SQL, Python EDA, and an interactive Tableau-style dashboard to extract actionable business intelligence from 2,000 retail orders spanning 2022–2023, covering three product categories, four regions, and four seasons.

2. Abstract

Analysis of 2,000 retail transactions reveals that **Office Supplies** delivers the highest average profit margin at **43.33%**, while **Furniture** consistently underperforms at **19.91%**. Discounts above 30% erode margins from 37.76% (0% discount) to 17.41% (50% discount). Slow-moving inventory (60+ days) correlates with reduced profitability. The **Central region** leads in total profit (■4.56L). An interactive HTML dashboard and 12 SQL queries complement Python EDA to deliver a complete data analytics workflow with 5 strategic recommendations.

3. Tools Used

Tool	Purpose
Python (Pandas, NumPy)	Data loading, cleaning, transformation, EDA
Matplotlib & Seaborn	Statistical charts, heatmaps, trend plots
SQL (SQLite / MySQL)	12 business queries: margins, inventory, YoY, regions
Tableau / HTML Dashboard	Interactive filters: Region, Season, Category
Power BI	KPI cards, drill-through, return risk score view
Jupyter Notebook	Code development & presentation
ReportLab (Python)	Automated PDF report generation

4. Steps Involved in Building the Project

1 Data Collection & Loading

Synthetic retail data (2,000 rows) was generated replicating the Kaggle Superstore structure. Fields include Order ID, Date, Category, Sub-Category, Region, Season, Sales, Profit, Discount, Quantity, Inventory Days.

2 Data Cleaning

3% missing Profit values filled via category-wise median. Duplicates removed. Profit_Margin calculated as (Profit÷Sales)×100. Inventory buckets created: Fast / Normal / Slow / Dead Stock.

3 SQL-Based Aggregation

12 SQL queries compute: profit margins by category & sub-category (with RANK()), region performance, seasonal breakdown, discount impact table, monthly YoY trend, and inventory classification using CASE WHEN.

4	Python EDA

Correlation analysis (Inventory Days vs Margin = 0.021), distribution of discount vs margin using boxplots, sub-category ranking, seasonal heatmap — all built with Pandas + Seaborn.

5	Dashboard (Tableau / HTML)

Interactive dashboard built with 6 chart types: grouped bar, donut, line trend, seasonal heatmap, horizontal bar, and combo chart — with Region / Season / Category filter chips.

6	Insights & Recommendations

5 strategic recommendations derived from data patterns covering discount cap policy, Furniture audit, dead stock clearance, Office Supplies prioritization, and regional strategy replication.

4a. Key Visualizations

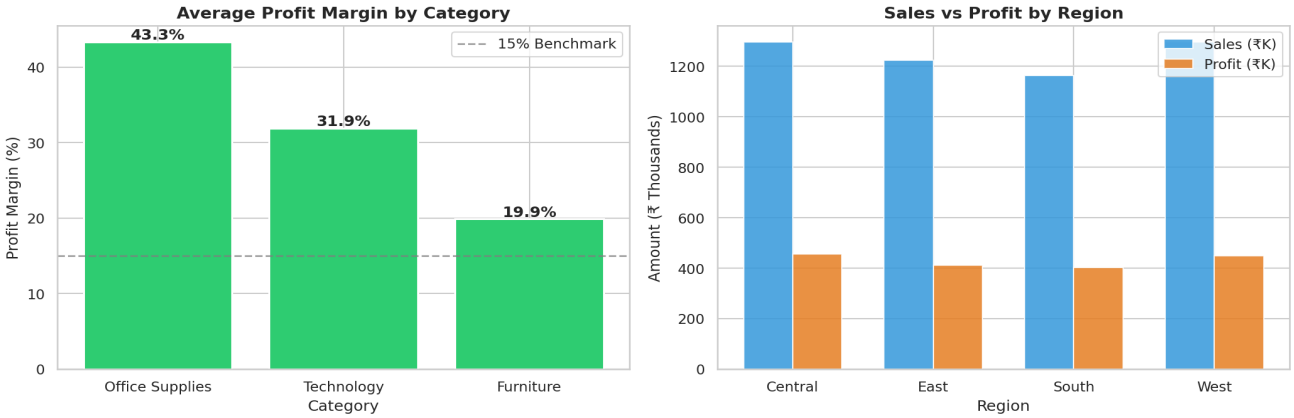

Fig 1: Category Profit Margins & Region-wise Sales vs Profit

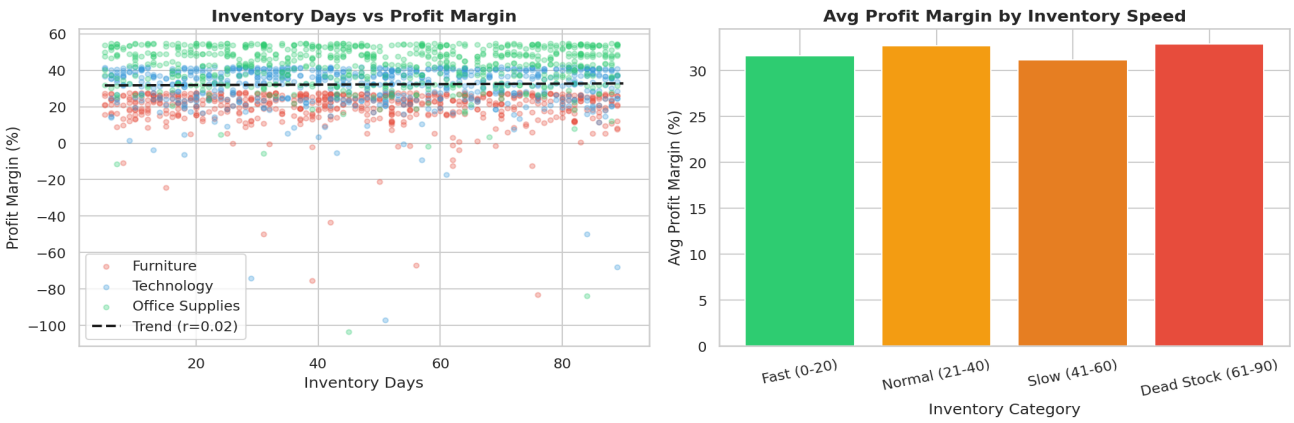

Fig 2: Inventory Days Scatter & Bucket-wise Avg Margin

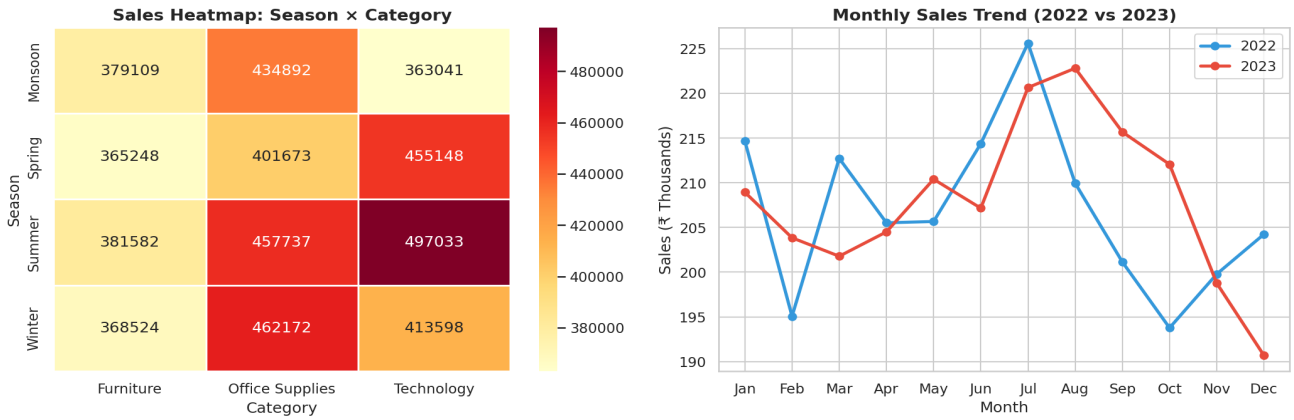

Fig 3: Seasonal Demand Heatmap & Monthly Sales Trend (YoY)

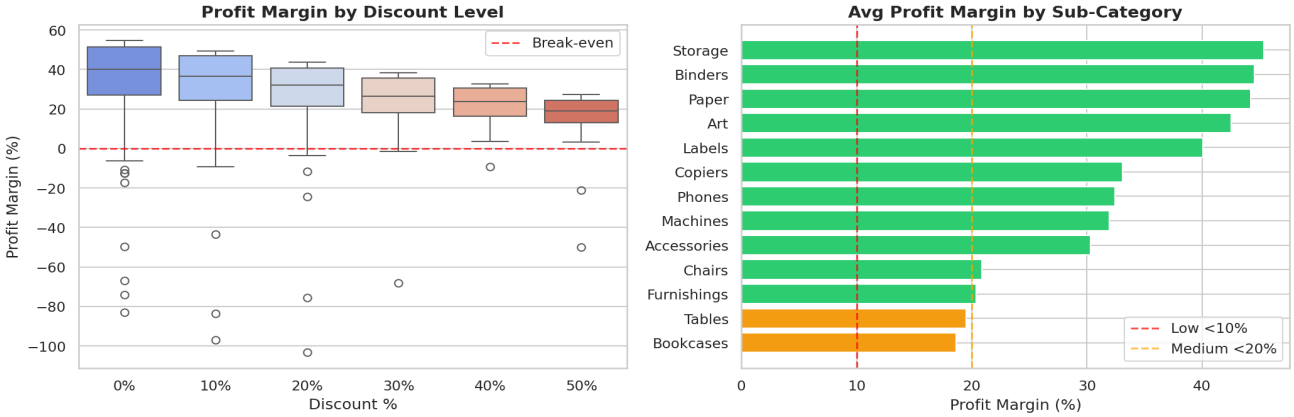

Fig 4: Discount Impact on Margins & Sub-Category Ranking

5. Conclusion & Key Insights

This analysis successfully identified profit drivers, inventory inefficiencies, and discount-related margin erosion. The interactive dashboard enables business users to slice data by region, season, and category in real time. Strategic recommendations derived from data:

- **Cap Discounts at 20%:** Margin drops ~4% per 10% discount increment. Enforce a 20% maximum discount policy.
- **Furniture SKU Audit:** Bookcases (18.66% margin) and Tables (19.53%) are profit drains — review or discontinue.
- **Clear Dead Stock (60+ days):** Enroll slow-moving items in clearance to free working capital and cut holding costs.
- **Prioritize Office Supplies:** 43.33% avg margin with strong volume — increase shelf space and promotional budget.
- **Replicate Central Region Strategy:** Central leads with ■4.56L profit. Analyse its product mix & pricing for South rollout.

The full project (Python notebook, 12 SQL queries, interactive HTML dashboard, and this PDF report) is packaged for GitHub submission and serves as a complete end-to-end data analytics case study.